



What does it take for  
a physical thing at  
one time to be the  
same as a physical  
thing at another  
time?

A solid blue circle with a subtle drop shadow, containing white text. The text is centered and reads: "Materialism:  
I am a wholly  
material  
thing".

**Materialism:**  
I am a wholly  
material  
thing



Simple  
dualism: I am  
an immaterial  
soul

**Fusion**  
dualism: I am  
a combination  
of a soul  
and a body

**Soul  
survival:**

X is me just in  
case X is the  
same immaterial  
soul as me

## Fusion survival:

X is me just in case X  
has the same immaterial  
soul as me and is the  
same material thing as  
me

How should a materialist respond?



The small changes view

X is the same material object as Y just in case X is causally connected to Y by a series of small physical changes over time.

It is very natural to respond by denying that the survival of a material thing requires that **all** of its parts continue to exist and be combined in the right way. Instead one might go for something like the following view of the survival of material things:





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the  $\mathcal{H}_\infty$  norm of the closed-loop system is bounded by  $\gamma$  and the  $\mathcal{H}_2$  norm of the closed-loop system is bounded by  $\beta$ .

Let  $\mathcal{H}_\infty$  and  $\mathcal{H}_2$  norms of the closed-loop system be denoted by  $\gamma$  and  $\beta$ , respectively. Then, the  $\mathcal{H}_\infty$  and  $\mathcal{H}_2$  norms of the closed-loop system are bounded by  $\gamma$  and  $\beta$ , respectively.

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The small changes view

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## The small changes view

X is the same material object as Y just in case X is causally connected to Y by a series of small physical changes over time.



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